Convergence of Riemannian Broyden Family of Quasi-Newton Methods with Wolfe Line Search

Xie Zhilin

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Abstract

We study the global convergence of a Riemannian Broyden family of quasi-Newton methods on complete Riemannian manifolds. The algorithm approximates the Riemannian Hessian via a convex combination of DFP and BFGS updates, transported between tangent spaces by an isometric vector transport satisfying a locking condition. This essay is mainly based on the Chapter 4 of the paper [**?**].

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1 Introduction and Preliminaries

Quasi-Newton methods are a class of optimization algorithms that approximate the Hessian matrix of the objective function to achieve faster convergence than gradient descent. Riemann Newton's method is a generalization of Newton's method to Riemannian manifolds, which allows us to optimize functions intrinsically defined on a Riemannian manifold. In this essay, we focus on the convergence analysis of Riemann Newton's method with Wolfe line search, which is a popular line search strategy that ensures sufficient decrease in the objective function.

1.1 Quasi-Newton Methods in Euclidean Space

Consider the unconstrained optimization problem in Euclidean space

$$\min_{x \in \mathbb{R}^n} f(x), \tag{1.1}$$

where $f : \mathbb{R}^n \to \mathbb{R}$ is a twice continuously differentiable function. Consider the Taylor expansion of its gradient at a point x^{k+1}

$$\nabla f(x) = \nabla f(x^{k+1}) + \nabla^2 f(x^{k+1})(x - x^{k+1}) + O(\|x - x^{k+1}\|^2)$$

denote $x = x^k, s^k = x^{k+1} - x^k$ and $y^k = \nabla f(x^{k+1}) - \nabla f(x^k)$, we have

$$y^k = \nabla^2 f(x^{k+1}) s^k + O(\|s^k\|^2)$$

if we avoid the $O(\|s^k\|^2)$ term, we have the *secant equation*

$$y^k = B^{k+1} s^k, \tag{1.2}$$

$$s^k = H^{k+1} y^k \tag{1.3}$$

the B^{k+1} and H^{k+1} are the approximations of the Hessian matrix and its inverse, respectively. We hope B^{k+1} is a positive definite matrix, that is

$$(s^k)^\top B^{k+1} s^k = (s^k)^\top y^k > 0 \tag{1.4}$$

for all $s^k \neq 0$. The condition (1.4) is called the *curvature condition.*

Definition 1.1 (Wolfe Conditions)**.** The condition

$$f(x^{k+1}) \leq f(x^k) + c_1 \nabla f(x^k)^\top s^k, \tag{1.5}$$
$$\nabla f(x^{k+1})^\top s^k \geq c_2 \nabla f(x^k)^\top s^k \tag{1.6}$$

for some constants $0 < c_1 < c_2 < 1$ are called the *Wolfe conditions.*

Lemma 1.1. *If the Wolfe conditions hold, then the curvature condition* (1.4) *holds.*

Proof. It is easy to check that

$$\nabla f(x^{k+1})^\top s^k \geq c_2 \nabla f(x^k)^\top s^k$$

we minus $\nabla f(x^k)^\top s^k$ on both sides, we have

$$(s^k)^\top y^k = \nabla f(x^{k+1})^\top s^k - \nabla f(x^k)^\top s^k \geq (c_2 - 1)\nabla f(x^k)^\top s^k$$

since $c_2 < 1$ and s^k is a descent direction, we have $\nabla f(x^k)^\top s^k < 0$, thus $(s^k)^\top y^k > 0$. $\square$

We have the *Davidon-Fletcher-Powell (DFP) update*

$$B^{k+1} = (I - \rho_k y^k (s^k)^\top) B^k (I - \rho_k s^k (y^k)^\top) + \rho_k y^k (y^k)^\top. \tag{1.7}$$

where $\rho_k = 1/(y^k)^\top s^k$.

Another famous update is the *Broyden-Fletcher-Goldfarb-Shanno (BFGS) update*

$$B^{k+1} = B^k + \frac{y^k (y^k)^\top}{(y^k)^\top s^k} - \frac{B^k s^k (s^k)^\top B^k}{(s^k)^\top B^k s^k} \tag{1.8}$$

As we know, the DFP formula (1.7) solves the following optimization problem

$$\begin{aligned} \min_B \quad & \|B - B^k\|_W \\ \text{s.t.} \quad & B = B^\top, \\ & Bs^k = y^k \end{aligned} \tag{1.9}$$

where W is an arbitrary matrix satisfying $Wy_k = s^k$ and $\|A\|_W = \|W^{1/2} A W^{1/2}\|_F$. A possible choice of W is the inverse of the average Hessian matrix, that is

$$W = \left(\int_0^1 \nabla^2 f(x^k + ts^k) dt \right)^{-1} \tag{1.10}$$

Similarly, the BFGS formula (1.8) solves the following optimization problem

$$\begin{aligned} \min_{H} \quad & \|H - H^k\|_W \\ \text{s.t.} \quad & H = H^\top, \\ & Hy^k = s^k \end{aligned} \tag{1.11}$$

Yuan proved that if the Wolfe conditions hold, then the DFP update is well-defined and H^{k+1} is positive definite.

Theorem 1.1 ([**?**]). *If the initial matrix H^0 is positive definite and the Wolfe conditions hold, the objective function f is twice continuously differentiable and bounded below, and the level set*

$$\mathcal{L} = \{x : f(x) \leq f(x^0)\}$$

is convex, and there exists a positive m and a positive M such that

$$m\|s\|^2 \leq s^\top \nabla^2 f(x) s \leq M\|s\|^2 \tag{1.12}$$

for all $s \in \mathbb{R}^n$ and $x \in \mathcal{L}$, then the DFP method with Wolfe line search converges globally to a critical point.

Broyden conbines the DFP update and the BFGS update, which is defined by

$$B^{k+1} = (1 - \phi_k) B^k_{\text{BFGS}} + \phi_k B^k_{\text{DFP}} \tag{1.13}$$

where $\phi_k \in [0, 1]$ is a parameter.

1.2 Some Notations and Definitions in Riemannian Geometry

1.2.1 Multilinear Algebra and Tensor

Definition 1.2. Assume R is a ring and V, W are two left R-modules. Define a equality relation $\sim$ on the free module generated by $V \times W$ as

$$\begin{aligned} (v_1 + v_2, w) &\sim (v_1, w) + (v_2, w), \\ (v, w_1 + w_2) &\sim (v, w_1) + (v, w_2), \\ (rv, w) &\sim r(v, w), \\ (v, rw) &\sim r(v, w) \end{aligned}$$

for all $v, v_1, v_2 \in V$, $w, w_1, w_2 \in W$ and $r \in R$. The quotient module is called the *tensor product* of V and W, denoted by $V \otimes_R W$ or simply $V \otimes W$ if there is no confusion. The equivalence class of (v, w) is denoted by $v \otimes w$.

If we do not specify the ring R, we assume R is a field, and the modules are vector spaces. The most important property of the tensor product is the following universal property.

Theorem 1.2 ([?] Theorem 18.3)**.** *Assume V, W, Z are R-modules, where R is commutative ring with identity. For a R-bilinear map $f : V \times W \to Z$, there exists a unique linear map $\tilde{f} : V \otimes_R W \to Z$, such that the following diagram commutes*

$$\begin{tikzcd} V \times W \arrow[r, "\otimes"] \arrow[dr, "f"'] & V \otimes_R W \arrow[d, "\tilde{f}"] \\ & Z \end{tikzcd}$$

if there exists another module T and $\phi : V \times W \to T$ is a R-bilinear map such that the following diagram commutes

$$\begin{tikzcd} V \times W \arrow[r, "\otimes"] \arrow[dr, "f"'] & T \arrow[d, "\tilde{f}"] \\ & Z \end{tikzcd}$$

then we have $T \cong V \otimes_R W$.

Using Theorem 1.2, we can derive the dual of tensor product.

Theorem 1.3 ([?] Proposition 18.15)**.** *Assyme V, W is free R-module with finite dimension, then there exists an R-module isomorphism*

$$\tilde{f} : V^* \otimes_R W^* \to (V \otimes_R W)^*,$$

such that for all $\alpha \in V^*, \beta \in W^*, v \in V, w \in W$, we have

$$\tilde{f}(\alpha \otimes \beta)(v \otimes w) = \alpha(v)\beta(w),$$

Theorem 1.4 ([?] Proposition 18.17)**.** *Let $f : V_1 \to V_2$ and $g : W_1 \to W_2$ is R-module homomorphism, then there exists unique R-module homomorphism*

$$f \otimes g : V_1 \otimes_R W_1 \to V_2 \otimes_R W_2,$$

such that for all $v \in V_1, w \in W_1$, we have

$$(f \otimes g)(v \otimes w) = f(v) \otimes g(w),$$

For an arbitrary F-linear space V, we define

$$\otimes^{l,k} V = \underbrace{V^* \otimes \cdots \otimes V^*}_{l \text{ times}} \otimes \underbrace{V \otimes \cdots \otimes V}_{k \text{ times}} \cong \mathrm{Hom}\left(\bigotimes^{l} V \otimes \bigotimes^{k} V^*, F \right) \tag{1.14}$$

1.2.2 Smooth Manifolds and Tangent Spaces

In brief, a smooth manifold is a topological space that locally looks like Euclidean space and has a smooth structure. The so-called smooth structure is to make sure the charts change smoothly, so that we can define the smoothness of functions on the manifold.

Definition 1.3 (Smooth Manifold)**.** A *smooth manifold* is a topological space M such that for each point $p \in M$, there exists an open neighborhood U of p and a homeomorphism $\varphi : U \to \mathbb{R}^n$. The pair (U, φ) is called a *chart* around p. A collection of charts $\{(U_\alpha, \varphi_\alpha)\}$ is called an *atlas* if $\cup_\alpha U_\alpha = M$ and the transition map $\varphi_\beta \circ \varphi_\alpha^{-1}$ is smooth for all α and β.

To derive smoothness of a function defined on a smooth manifold, we need to use the charts to pull back and push forward the function to Euclidean space.

Definition 1.4 (Smooth Map)**.** Let M^m and N^n are two smooth manifolds. A map $f : M \to N$ is called *smooth* if for any charts (U, φ) around $p \in M$ and (V, ψ) around $f(p) \in N$, the map $\psi \circ f \circ \varphi^{-1}$ is smooth from $\varphi(U) \subseteq \mathbb{R}^m$ to $\psi(V) \subseteq \mathbb{R}^n$.

Definition 1.5 (Tangent Space)**.** Let M is a smooth manifold and $p \in M$ is a point. A *tangent vector* is a linear map $v_p : C^\infty(M) \to \mathbb{R}$ such that for all $f, g \in C^\infty(M)$ it satisfies the Leibniz rule

$$v_p(fg) = f(p)v_p(g) + g(p)v_p(f)$$

The set of all tangent vectors at p is called the *tangent space* at p, denoted by T_pM.

A basic example of tangent vector is derived by charts. Let (U, φ) is a chart around p, we can define the tangent vector ∂_i by

$$\partial_i(f) = \frac{\partial(f \circ \varphi^{-1})}{\partial x_i}(\varphi(p))$$

where $\frac{\partial}{\partial x_i}$ is the partial derivative in Euclidean space. It is easy to check that $\{\partial_1, \cdots, \partial_n\}$ is a basis of T_pM. Thus we can identify T_pM with $\mathbb{R}^n$ by the chart (U, φ).

Denote the (k, l)-tensor of T_pM by $\otimes^{k,l}T_pM$.

Definition 1.6 (Tensor Field)**.** A *tensor field* of type (k, l) on M is a map

$$T : M \to \cup_{p \in M} \otimes^{k,l} T_pM$$

such that for each $p \in M$, $T(p) \in \otimes^{k,l}T_pM$ and for any smooth vector fields $X_1, \cdots, X_k$ and any smooth covector fields $\omega_1, \cdots, \omega_l$, the function

$$T(X_1, \cdots, X_k, \omega_1, \cdots, \omega_l) : M \to \mathbb{R}$$

defined by

$$T(X_1, \cdots, X_k, \omega_1, \cdots, \omega_l)(p) = T(p)(X_1(p), \cdots, X_k(p), \omega_1(p), \cdots, \omega_l(p))$$

is smooth.

1.2.3 Riemann Manifolds and Riemannian Metrics

Definition 1.7 (Riemannian Metric)**.** Let M is a smooth manifold. A *Riemannian metric* on M is a second order covariant tensor field g such that

(a) For each $p \in M$, the map $g_p : T_pM \times T_pM \to \mathbb{R}$ is a positive definite inner product.

(b) For any smooth vector fields X and Y, the function $g(X, Y) : M \to \mathbb{R}$ defined by $g(X, Y)(p) = g_p(X(p), Y(p))$ is smooth.

given a Riemannian metric g, we call the pair (M, g) a *Riemannian manifold.*

By partition of unity, we can always construct a Riemannian metric on a second countable smooth manifold.

Theorem 1.5 (Fundamental Theorem of Riemannian Geometry)**.** *Let M be a second countable smooth manifold, then there always exists a Riemann metric on M.*

1.2.4 Covariant Derivative, Levi-Civita Connection, Parallel Transport and Geodesic

In order to generalize derivatives to Riemannian manifolds, we need to introduce the notion of covariant derivative.

A covariant derivative is a way to differentiate vector fields along other vector fields on a Riemannian manifold. Let (M, g) is a Riemannian manifold, a *affine connection* is a map $\nabla : \Gamma^\infty(TM) \times \Gamma^\infty(TM) \to \Gamma^\infty(TM)$, where $\Gamma^\infty(TM)$ is the set of all smooth vector fields on M, such that for all $X, Y, Z \in \Gamma^\infty(TM)$ and $f \in C^\infty(M)$, we have

(a) Smooth function Linearity. $\nabla_{fX+gY} Z = f\nabla_X Z + g\nabla_Y Z$ for all $f, g \in C^\infty(M)$.

(b) Linearity. $\nabla_X(aY + bZ) = a\nabla_X Y + b\nabla_X Z$ for all real numbers a, b.

(c) Leibniz rule. $\nabla_X(fY) = f\nabla_X Y + X(f)Y$ for all $f \in C^\infty(M)$.

Jost gives an explanation to the name "affine connection" in [?]. The symbol $\nabla_X Y$ is called the *covariant derivative* of Y along X. The covariant derivative $\nabla_X Y$ can be regarded as the directional derivative of Y along X.

A covariant derivative ∇ can be regarded as a $(1,2)$-tensor field.

We called a covariant derivative ∇ is *compatible* with the Riemannian metric g if for all $X, Y, Z \in \Gamma^\infty(TM)$, we have

$$X(g(Y,Z)) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$$

A covariant derivative ∇ is *torsion-free* if for all $X, Y \in \Gamma^\infty(TM)$, we have

$$\nabla_X Y - \nabla_Y X = [X, Y]$$

Theorem 1.6 (Levi-Civita Connection). *Let (M, g) is a Riemannian manifold, then there exists a unique covariant derivative ∇ that is compatible with g and torsion-free. We call ∇ the* Levi-Civita connection *of (M, g).*

Given a affine connection ∇, we can generalize it to all type of tensor fields by the following rules.

Theorem 1.7 ([?] Theorem 3.2.1). *Assume ∇ is a affine connection on a Riemannian manifold (M, g). For a fixed vector field X, we can define the covariant derivative of arbitrary tensor field along X denote by $\nabla_X : \Gamma^\infty(\otimes^{k,l}TM) \to \Gamma^\infty(\otimes^{k,l}TM)$ by the following rules*

(a) *For all type of tensor fields θ and η, we have*

$$\nabla_X(\theta \otimes \eta) = \nabla_X \theta \otimes \eta + \theta \otimes \nabla_X \eta$$

(b) *For smooth function f and g, tensor field θ, vector field X and Y, we have*

$$\nabla_{fX+gY}\theta = f\nabla_X\theta + g\nabla_Y\theta$$

(c) *For real number a and b, tensor field θ and η, vector field X, we have*

$$\nabla_X(a\theta + b\eta) = a\nabla_X\theta + b\nabla_X\eta$$

the formula of such ∇_X is given by

$$\begin{aligned}\nabla_X\left(T(Y_1, \ldots, Y_k, \omega_1, \ldots, \omega_l)\right) &= (\nabla_X T)(Y_1, \ldots, Y_k, \omega_1, \ldots, \omega_l) \\ &+ \sum_{i=1}^{k} T(Y_1, \ldots, \nabla_X Y_i, \ldots, Y_k, \omega_1, \ldots, \omega_l) \\ &+ \sum_{j=1}^{l} T(Y_1, \ldots, Y_k, \omega_1, \ldots, \nabla_X \omega_j, \ldots, \omega_l).\end{aligned}$$

Given a smooth curve $\gamma : [0,1] \to M$ on a Riemannian manifold (M,g), we can define the covariant derivative along γ by $\nabla_{\dot\gamma} = \nabla_{\dot\gamma(t)}$. A vector field $Y = Y^i \partial_i$ along γ is called a *parallel vector field* if $\nabla_{\dot\gamma} Y = 0$, that is

$$\nabla_{\dot\gamma} Y = \frac{dx^i}{dt}\left(\frac{dY^k}{dx^i} + Y^j \Gamma_{ij}^k\right)\partial_k = \left(\frac{dY^k}{dt} + \frac{dx^i}{dt} Y^j \Gamma_{ij}^k\right)\partial_k = 0.$$

which is called the *parallel transport equation.* A curve γ is called a *geodesic* if $\nabla_{\dot\gamma}\dot\gamma = 0$, that is

$$\nabla_{\dot\gamma}\dot\gamma = \left(\frac{d^2\gamma^k}{dt^2} + \Gamma_{ij}^k \frac{d\gamma^i}{dt}\frac{d\gamma^j}{dt}\right)\partial_k = 0.$$

1.2.5 Retraction and Vector Transport

Recall that an exponential map $\exp_p : T_pM \to M$ is defined by $\exp_p(v) = \gamma_v(1)$, where γ_v is the geodesic starting at p with initial velocity v. We generalize the exponential map to a retraction since exponential map may not be easy to compute in practice.

Definition 1.8 (Retraction)**.** Let M is a smooth manifold. A *retraction* is a smooth map

$$R : TM \to M, \quad (p,v) \mapsto R_p(v)$$

such that for each curve $c(t) = R_p(tv)$, we have $c(0) = p$ and $c'(0) = v$.

We notice that $dR_p(0) = \mathrm{id}$, where $dR_p(0)$ is the differential of R_p at 0. By the inverse function theorem, there exists a neighborhood U of 0 such that R_p is a diffeomorphism from U to $R_p(U)$. Thus we can use the retraction to define a local chart around a fixed p. This fact gives us a way to compute the Riemannian gradient.

Proposition 1.1. *Let $f : M \to \mathbb{R}$ be a smooth function on a Riemannian manifold (M,g) equipped with a retraction R. For any fixed point $p \in M$, we have*

$$\operatorname{grad} f(p) = \operatorname{grad}(f \circ R_p)(0)$$

Proof. By the chain rule, we have

$$d(f \circ R_p)(0)[v] = df(p)[dR_p(0)[v]] = df(p)[v] = \langle \operatorname{grad} f(p), v\rangle_p$$

since gradient is unique to

$$\langle \operatorname{grad}(f \circ R_p)(0), v\rangle_p = d(f \circ R_p)(0)[v] = \langle \operatorname{grad} f(p), v\rangle_p$$

for all $v \in T_pM$, we have $\operatorname{grad} f(p) = \operatorname{grad}(f \circ R_p)(0)$. $\square$

Parallel transport is a way to transport a tangent vector along a curve while keeping it parallel to itself. Let $\gamma : [0, 1] \to M$ is a curve on a Riemannian manifold (M, g) and denote ∇ the Levi-Civita connection, we say a vector field V along γ is parallel if $\nabla_{\dot\gamma} V = 0$. The parallel transport of a vector $v \in T_{\gamma(0)}M$ along γ is the unique parallel vector field V along γ such that $V(0) = v$. We denote the parallel transport by $P^{\gamma}_{a\to b} : T_{\gamma(a)}M \to T_{\gamma(b)}M$.

Proposition 1.2. *Parallel transport is an isometry.*

Proof. Let X and Y are two parallel vector fields along γ, since ∇ is compatible with the Riemannian metric, we have

$$\frac{d}{dt}\langle X, Y\rangle = \langle \nabla_{\dot\gamma} X, Y\rangle + \langle X, \nabla_{\dot\gamma} Y\rangle = 0$$

□

We generalize the parallel transport to a vector transport.

Definition 1.9 (Vector Transport)**.** Let M is a smooth manifold. A *vector transport* is a smooth map

$$\mathcal{T} : TM \oplus TM \to TM, \quad (p, v, w) \mapsto \mathcal{T}_{v_p}(w)$$

such that for some retraction R we have

(a) For fixed p and v, the map $\mathcal{T}_{v_p} : T_pM \to T_{R_p(v)}M$ is a linear map.

(b) For all $p \in M$ we have that $\mathcal{T}_{0_p}$ is the identity map on T_pM.

Here is a direct comprehension of the vector transport. It transports a tangent vector w at p to a tangent vector at $R_p(v)$, and the transport is linear in w.

There are two kinds of vector transport we usually use beside parallel transport. The first one is the differentiated retraction, which satisfies

$$dR_p(v)[w_p] = \mathcal{T}_{v_p}(w_p)$$

for all $p \in M$ and $v, w \in T_pM$, and we denote the vector transport $\mathcal{T}_R$. The second one is the *isometric* vector transport, which satisfies

$$g_{R_p(v)}(\mathcal{T}_{v_p}(w), \mathcal{T}_{v_p}(w)) = g_p(w, w)$$

for all $p \in M$ and $v, w \in T_pM$, and we denote the vector transport $\mathcal{T}_S$.

2 Frame of Riemannian Broyden's Method

2.1 Secant Equation on Riemannian Manifolds

The following equation is a generalization of Taylor expansion of a vector field X at a point p to a point q.

Theorem 2.1. *Let (M, g) be a Riemannian manifold and R is a retraction on M. Let X is a smooth vector field on M. Consider $\gamma : [0, 1] \to M$ is geodesic with initial velocity $\dot{\gamma}(0) = v$ and $\gamma(0) = p$, we have*

$$P^{\gamma}_{1\to 0} X_{\gamma(1)} = X_p + \sum_{i=1}^{k-1} \frac{1}{i!} \nabla^i_{\dot{\gamma}} X_p(v, \cdots, v) + R_k \tag{2.1}$$

where R_k is the remainder term

$$R_k = \int_0^1 \frac{(1-t)^{k-1}}{(k-1)!} P^{\gamma}_{t\to 0} \left(\nabla^k X_{\gamma(t)} \underbrace{(\dot{\gamma}(t), \cdots, \dot{\gamma}(t))}_{k \ times} \right) dt$$

Proof. The proof is similar to the Taylor expansion of a function in Euclidean space. The only thing we need to do is to use parallel transport to pullback the vector field to the tangent space at p and use the covariant derivative instead of the usual derivative.

We define

$$Y(t) = P^{\gamma}_{t\to 0} X_{\gamma(t)}$$

then $Y : [0, 1] \to T_pM$ is a curve in the tangent space at p. Let $\{e_i\}_{1\le i\le n}$ is an orthonormal basis of T_pM, and $\{E_i(t)\}_{1\le i\le n}$ is the parallel transport of $\{e_i\}_{1\le i\le n}$ along γ, then $\{E_i(t)\}_{1\le i\le n}$ is an orthonormal basis of $T_{\gamma(t)}M$. We can write $X_{\gamma(t)}$ as

$$X_{\gamma(t)} = X^i(t) E_i(t)$$

here we use Einstein summation convention. Since $E_i(t)$ is parallel along γ, we have

$$\nabla_{\dot{\gamma}} X_{\gamma(t)} = \dot{X}^i(t) E_i(t) + X^i(t) \nabla_{\dot{\gamma}} E_i(t) = \dot{X}^i(t) E_i(t) \tag{2.2}$$

thus

$$Y(t) = P^{\gamma}_{t\to 0} \left(X^i(t) E_i(t) \right) = X^i(t) e_i \tag{2.3}$$

now that $Y(t)$ is a curve in the vector space T_pM which is isomorphic to $\mathbb{R}^n$, we can use the Taylor expansion of a function in Euclidean space to write $Y(t)$ as

$$\dot{Y}(t) = \dot{X}^i(t) e_i = P_{t\to 0} \left(\dot{X}^i(t) E_i(t) \right) = P^{\gamma}_{t\to 0} \left(\nabla_{\dot{\gamma}} X_{\gamma(t)} \right) \tag{2.4}$$

the last equality is from (2.2). We can repeat the above process to get

$$Y^{(i)}(t) = P^{\gamma}_{t\to 0}\left(\nabla^i X_{\gamma(t)} \underbrace{(\dot\gamma(t), \cdots, \dot\gamma(t))}_{k \text{ times}}\right) \tag{2.5}$$

since $Y(t)$ is a curve in the vector space T_pM which is isomorphic to $\mathbb{R}^n$, we can use the Taylor expansion of a function in Euclidean space to write $Y(t)$ as

$$Y(t) = Y(0) + \sum_{i=1}^{k-1} \frac{t^i}{i!} Y^{(i)}(0) + R_k \tag{2.6}$$

where the remainder term R_k is given by

$$R_k = \int_0^1 \frac{(1-t)^{k-1}}{(k-1)!} Y^{(k)}(t)dt = \int_0^1 \frac{(1-t)^{k-1}}{(k-1)!} P^{\gamma}_{t\to 0}\left(\nabla^k_{\dot\gamma} X_{\gamma(t)}\right) dt \tag{2.7}$$

combining (2.5), (2.6) and (2.7), and $\dot\gamma(0) = v$, we have

$$P^{\gamma}_{1\to 0} X_{\gamma(1)} = Y(1) = Y(0) + \sum_{i=1}^{k-1} \frac{1}{i!} Y^{(i)}(0) + R_k = X_p + \sum_{i=1}^{k-1} \frac{1}{i!} \nabla^i_{\dot\gamma} X_p \underbrace{(v, \cdots, v)}_{i \text{ times}} + R_k$$

□

Remark 2.1. Since γ is a geodesic, we have

$$\nabla^2 X_{\gamma(t)}(\dot\gamma(t), \dot\gamma(t)) = \nabla_{\dot\gamma}\left(\nabla_{\dot\gamma} X\right)_{\gamma(t)} - \nabla_{\nabla_{\dot\gamma}\dot\gamma} X_{\gamma(t)} = \nabla_{\dot\gamma}\left(\nabla_{\dot\gamma} X\right)_{\gamma(t)}$$

the higher order covariant derivative can be computed by

$$\nabla^i X_{\gamma(t)}(\dot\gamma(t), \cdots, \dot\gamma(t)) = \left(\underbrace{\nabla_{\dot\gamma}\nabla_{\dot\gamma}\left(\cdots \nabla_{\dot\gamma} X\right)}_{i \text{ times}}\right)_{\gamma(t)}$$

Similar to Quasi-Newton's method in Euclidean space, we can use the Taylor expansion of the Riemannian gradient to derive the secant equation on Riemannian manifolds.

Let $f : M \to \mathbb{R}$ is a smooth function on a Riemannian manifold (M, g), and the curve $\gamma_k : [0, 1] \to M$ is a geodesic that connects x^k and x^{k+1}. Denote the initial velocity of γ_k by s_0^k, we have

$$s_0^k = \exp^{-1}_{x^k}(x^{k+1}) = \dot\gamma_k(0)$$

if we avoid the remainder term, we have

$$\begin{aligned} P^{\gamma_k}_{1\to 0} \operatorname{grad} f(x^{k+1}) &= \operatorname{grad} f(x^k) + \nabla_{s_0^k} \operatorname{grad} f(x^k) + R \\ &= \operatorname{grad} f(x^k) + \operatorname{Hess} f(x^k)[s_0^k] + R \end{aligned}$$

where R is the remainder term. But the left term is in $T_{x^k}M$, we use the parallel transport to push it to $T_{x^{k+1}}M$, we have

$$\operatorname{grad} f(x^{k+1}) = P_{0\to 1}^{\gamma_k}\operatorname{grad} f(x^k) + P_{0\to 1}^{\gamma_k}\left(\operatorname{Hess} f(x^k)[s_0^k]\right) + R$$

finally we avoid R and we have the secant equation on Riemannian manifolds

$$\operatorname{grad} f(x^{k+1}) = P_{0\to 1}^{\gamma_k}\operatorname{grad} f(x^k) + \mathcal{B}^{k+1}[P_{0\to 1}^{\gamma_k}(s_0^k)] \tag{2.8}$$

if we use y^k to denote $\operatorname{grad} f(x^{k+1}) - P_{0\to 1}^{\gamma_k}\operatorname{grad} f(x^k)$ and s^k to denote $P_{0\to 1}^{\gamma_k}(s_0^k)$, we have

$$y^k = \mathcal{B}^{k+1}s^k \tag{2.9}$$

which is the secant equation on Riemannian manifolds. Similar to the Euclidean case, we hope $\mathcal{B}^{k+1}$ is a positive definite operator, that is

$$g_{x^{k+1}}(s^k, y^k) > 0 \tag{2.10}$$

which is the curvature condition on Riemannian manifolds.

2.2 Secant Equation with General Retraction and Vector Transport

It is worth noting that all the above discussion is based on the geodesic γ_k that connects x^k and x^{k+1}, in other words, we use the exponential map to define the secant equation on Riemannian manifolds. We can also use a general retraction to define the secant equation on Riemannian manifolds, which is given by

$$y^k = \mathcal{B}^{k+1}s^k \tag{2.11}$$

where $s^k = P_{0\to 1}^{\gamma_k}(R_{x^k}^{-1}(x^{k+1}))$. The only possible issue is that in the second order term

$$\ddot{Y}(t) = P_{t\to 0}^{\gamma}\left(\nabla_{\dot\gamma}\left(\nabla_{\dot\gamma}X\right)_{\gamma(t)}\right) = P_{t\to 0}^{\gamma}\left(\nabla_{\dot\gamma}^2 X_{\gamma(t)} + \nabla_{\nabla_{\dot\gamma}\dot\gamma}X_{\gamma(t)}\right)$$

when $t = 0$, it turns to

$$\ddot{Y}(0) = \nabla_{\dot\gamma}^2 X_p + \nabla_{\nabla_{\dot\gamma}\dot\gamma}X_p$$

therefore, when we use a general retraction to define the secant equation on Riemannian manifolds, we want to let the extra term $\nabla_{\nabla_{\dot\gamma}\dot\gamma}X_p$ vanish, then everything will be the same as the case of using the exponential map.

2.3 Broyden's Update on Riemannian Manifolds

As in the Euclidean space the matrix B^{k+1} solves the optimization problem (1.9), we can also define the DFP update on Riemannian manifolds by solving the following optimization problem

$$\begin{aligned} \min_{\mathcal{B}} \quad & \|\mathcal{B} - \mathcal{B}^k\|_W \\ \text{s.t.} \quad & \mathcal{B} = \mathcal{B}^*, \\ & \mathcal{B}s^k = y^k \end{aligned} \tag{2.12}$$

where $\|\mathcal{A}\|_W = \|W^{1/2}G^{1/2}\mathcal{A}G^{1/2}W^{1/2}\|_F$ and G is the matrix representation of the Riemannian metric defined by $G_{ij} = g_{ij}$. We then have the following DFP update on Riemannian manifolds

$$\mathcal{B}^{k+1} = \left(\mathrm{id} - \frac{y^k(s^k)^\flat}{g(y_k, s_k)}\right)\tilde{\mathcal{B}}^k\left(\mathrm{id} - \frac{s^k(y^k)^\flat}{g(y_k, s_k)}\right) + \frac{y^k(y^k)^\flat}{g(y_k, s_k)} \tag{2.13}$$

where $\flat : T_pM \to T_p^*M$ is the musical isomorphism defined by $v^\flat(w) = g_p(v, w)$ for all $v, w \in T_pM$, and $\tilde{\mathcal{B}}^k$ is defined as

$$\tilde{\mathcal{B}}^k = P_{0\to 1}^{\gamma_k} \circ \mathcal{B}^k \circ P_{1\to 0}^{\gamma_k}$$

the parallel transport can be replaced by any isometric vector transport such as

$$\tilde{\mathcal{B}}^k = \mathcal{T}_{S_{\alpha^k\eta^k}} \circ \mathcal{B}^k \circ \mathcal{T}_{S_{\alpha^k\theta^k}}^{-1} \tag{2.14}$$

where $\mathcal{B}^k\eta^k = -\mathrm{grad}\, f(x^k)$ and $x^{k+1} = R_{x^k}(\alpha^k\eta^k)$. We do this transform is to map $\mathcal{B}^k$ from $T_{x^k}M$ to $T_{x^{k+1}}M$, and the isometry of the vector transport ensures that $\tilde{\mathcal{B}}^k$ is positive definite if $\mathcal{B}^k$ is positive definite, that is, the curvature condition is preserved.

Similarly, we can also get the BFGS update on Riemannian manifolds

$$\mathcal{B}^{k+1} = \tilde{\mathcal{B}}^k + \frac{y^k(y^k)^\flat}{g(y_k, s_k)} - \frac{\tilde{\mathcal{B}}^k s^k(\tilde{\mathcal{B}}^k s^k)^\flat}{g(s_k, \tilde{\mathcal{B}}^k s^k)} \tag{2.15}$$

Then for any $\phi_k \in [0, 1]$, we can combine the DFP update and the BFGS update to get the Broyden's update on Riemannian manifolds

$$\begin{aligned} \mathcal{B}^{k+1} &= (1 - \phi_k)\mathcal{B}^k_{\mathrm{BFGS}} + \phi_k\mathcal{B}^k_{\mathrm{DFP}} \\ \mathcal{B}^{k+1}(\cdot) &= \tilde{\mathcal{B}}^k(\cdot) + \frac{y^k(y^k)^\flat(\cdot)}{g(y_k, s_k)} - \frac{\tilde{\mathcal{B}}^k s^k(\tilde{\mathcal{B}}^k s^k)^\flat(\cdot)}{g(s_k, \tilde{\mathcal{B}}^k s^k)} + \phi_k g(s^k, \tilde{\mathcal{B}}^k s^k)v^k(v^k)^\flat(\cdot) \end{aligned} \tag{2.16}$$

where

$$v^k = \frac{y^k}{g(y^k, s^k)} - \frac{\tilde{\mathcal{B}}^k s^k}{g(s^k, \tilde{\mathcal{B}}^k s^k)}$$

we usually choose ϕ_k in the interval (ϕ_k^c, ∞) where ϕ_k^c is given by

$$\begin{aligned} \phi_k^c &= \frac{1}{1-\mu^k} \\ \mu^k &= \left(\frac{1}{g(y^k, s^k)}\right)^2 g(y^k, (\tilde{\mathcal{B}}^k)^{-1} y^k) g(s^k, \tilde{\mathcal{B}}^k s^k) \end{aligned} \tag{2.17}$$

2.4 Wolfe Line Search on Riemannian Manifolds

The Euclidean Wolfe conditions (1.5) and (1.6) can be easily generalized to Riemannian manifolds by replacing the Euclidean inner product with the Riemannian metric. Let $\mathcal{B}^k \eta^k = -\operatorname{grad} f(x^k)$ and $x^{k+1} = R_{x^k}(\alpha^k \eta^k)$. Fix any $0 < c_1 < c_2 < 1$, the Wolfe conditions on Riemannian manifolds are given by

$$f(x^{k+1}) \leq f(x^k) + c_1 \alpha^k g_{x^k}(\operatorname{grad} f(x^k), \eta^k), \tag{2.18}$$

$$\frac{d}{dt} f\left(R_{x^k}(t\eta^k)\right)\Big|_{t=\alpha^k} \geq c_2 \frac{d}{dt} f\left(R_{x^k}(t\eta^k)\right)\Big|_{t=0} \tag{2.19}$$

2.5 Locking Condition

We assume $\mathcal{T}_R$ and $\mathcal{T}_S$ satisfy

$$\|\mathcal{T}_{R_\eta} - \mathcal{T}_{S_\eta}\|_p \leq \delta \|\eta\|, \tag{2.20}$$

$$\|\mathcal{T}_{R_\eta}^{-1} - \mathcal{T}_{S_\eta}^{-1}\|_p \leq \delta \|\eta\|, \tag{2.21}$$

for all $p \in M$ and $\eta \in T_p M$, where δ is a fixed constant. We then have the following locking condition

$$\begin{aligned} \mathcal{T}_{S_\eta} \eta &= \beta \mathcal{T}_{R_\eta} \eta \\ \beta &= \frac{\|\eta\|}{\|\mathcal{T}_{R_\eta} \eta\|} \end{aligned} \tag{2.22}$$

if locking condition is satisfied, we have $\mathcal{T}_{S_\eta} \eta = \beta \mathcal{T}_{R_\eta} \eta$, and

$$g(\mathcal{T}_{S_\eta} \eta, \mathcal{T}_{S_\eta} \eta) = \beta^2 g(\mathcal{T}_{R_\eta} \eta, \mathcal{T}_{R_\eta} \eta) = g(\eta, \eta) \quad = \beta^2 g(\eta, \eta) \tag{2.23}$$

which assures the curvature condition is preserved when we use $\mathcal{T}_S$ to push out $\mathcal{B}^k$.

2.6 The Frame of Riemannian Broyden's Method

The Riemannian Broyden's method is given by the following algorithm.

Algorithm 2.1 Riemannian Broyden's Method

Input: A Riemannian manifold (M, g), a retraction R, an isometric vector transport $\mathcal{T}_S$ with R as associated retraction, a second differentiable function $f : M \to \mathbb{R}$, an initial point $x^0 \in M$, an initial positive definite operator $\mathcal{B}^0 : T_{x_0}M \to T_{x_0}M$, a sequence $\{\phi_k\}_{k\geq 0}$ in $[0, 1]$, Wolfe line search parameters $0 < c_1 < c_2 < 1$ and a tolerance $\epsilon > 0$.

1. $k \leftarrow 1$;
2. **while** $\|\text{grad}\, f(x^k)\|_{x^k} > \epsilon$ **do**
3. Compute search direction $\eta^k \leftarrow -(\mathcal{B}^k)^{-1}\text{grad}\, f(x^k)$;
4. Set $\alpha^k \leftarrow$ as step size that satisfies the Wolfe conditions (2.18) and (2.19)

$$f(x^{k+1}) \leq f(x^k) + c_1\alpha^k g_{x^k}(\text{grad}\, f(x^k), \eta^k),$$
$$\frac{d}{dt} f\left(R_{x^k}(t\eta^k)\right)\Big|_{t=\alpha^k} \geq c_2 \frac{d}{dt} f\left(R_{x^k}(t\eta^k)\right)\Big|_{t=0}$$

5. Update $x^{k+1} \leftarrow R_{x^k}(\alpha^k\eta^k)$;
6. Let $s^k \leftarrow \mathcal{T}_{S_{\alpha^k\eta^k}}(\alpha^k\eta^k)$ and $y^k \leftarrow (\beta^k)^{-1}\text{grad}\, f(x^{k+1}) - \mathcal{T}_{S_{\alpha^k\eta^k}}(\text{grad}\, f(x^k))$, where

$$\beta^k = \frac{\|\alpha_k\eta_k\|_{x^k}}{\|\mathcal{T}_{R_{\alpha^k\eta^k}}(\alpha^k\eta^k)\|_{x^{k+1}}}$$

   and $\mathcal{T}_{R_{\alpha^k\eta^k}}$ is the differentiated retraction vector transport;
7. Push out $\tilde{\mathcal{B}}^k \leftarrow \mathcal{T}_{S_{\alpha^k\eta^k}} \circ \mathcal{B}^k \circ \mathcal{T}^{-1}_{S_{\alpha^k\eta^k}}$;
8. Compute $\mathcal{B}^{k+1} \leftarrow (1 - \phi_k)\mathcal{B}^k_{\text{BFGS}} + \phi_k\mathcal{B}^k_{\text{DFP}}$ where $\mathcal{B}^k_{\text{BFGS}}$ and $\mathcal{B}^k_{\text{DFP}}$ are given by (2.15) and (2.13) respectively, that is

$$\mathcal{B}^{k+1} = (1 - \phi_k)\mathcal{B}^k_{\text{BFGS}} + \phi_k\mathcal{B}^k_{\text{DFP}}$$
$$\mathcal{B}^{k+1}(\cdot) = \tilde{\mathcal{B}}^k(\cdot) + \frac{y^k(y^k)^\flat(\cdot)}{g(y_k, s_k)} - \frac{\tilde{\mathcal{B}}^k s^k(\tilde{\mathcal{B}}^k s^k)^\flat(\cdot)}{g(s_k, \tilde{\mathcal{B}}^k s^k)} + \phi_k g(s^k, \tilde{\mathcal{B}}^k s^k)v^k(v^k)^\flat(\cdot)$$

   where

$$v^k = \frac{y^k}{g(y^k, s^k)} - \frac{\tilde{\mathcal{B}}^k s^k}{g(s^k, \tilde{\mathcal{B}}^k s^k)}$$

   and ϕ^k is a fixed parameter in the interval (ϕ^c_k, ∞) where ϕ^c_k is given by

$$\phi^c_k = \frac{1}{1 - \mu^k}$$
$$\mu^k = \left(\frac{1}{g(y^k, s^k)}\right)^2 g(y^k, (\tilde{\mathcal{B}}^k)^{-1}y^k)g(s^k, \tilde{\mathcal{B}}^k s^k)$$

and

$$\tilde{\mathcal{B}}^k = \mathcal{T}_{S_{\alpha^k \eta^k}} \circ \mathcal{B}^k \circ \mathcal{T}^{-1}_{S_{\alpha^k \eta^k}}$$

9. $\quad k \leftarrow k + 1$;

10. **end while**

3 Convergence Analysis

3.1 Locking Condition

The following lemma shows that under the locking condition, the curvature condition is preserved under Wolfe line search.

Lemma 3.1 (Sylvester)**.**

Lemma 3.2. *The algorithm 2.1 will not stop unless the stopping criterion is satisfied. For all integer $k \geq 0$, the Hessian approximation $\mathcal{B}^k$ is self-adjoint and positive definite, and for $\eta^k \neq 0$ we have*

$$g(s^k, y^k) \geq (c_2 - 1)\alpha^k g(\operatorname{grad} f(x^k), \eta^k) > 0 \tag{3.1}$$

Proof. If all quantities exist and $\eta^k \neq 0$, define $\widetilde{m}_k = f\left(R_{x^k}(t\eta^k/\|\eta^k\|)\right)$, we notice that

$$\begin{aligned} \frac{d}{dt}\widetilde{m}_k(t) &= \frac{d}{dt} f\left(R_{x^k}\left(\frac{t\eta^k}{\|\eta^k\|}\right)\right) \\ &= g_{R_{x^k}(t\eta^k/\|\eta^k\|)}\left(\operatorname{grad} f\left(R_{x^k}\left(\frac{t\eta^k}{\|\eta^k\|}\right)\right), \mathcal{T}^R_{t\eta^k/\|\eta^k\|}\left(\frac{\eta^k}{\|\eta^k\|}\right)\right) \end{aligned} \tag{3.2}$$

then

$$\begin{aligned} g(s^k, y^k) &= g\left(\mathcal{T}_{S_{\alpha^k\eta^k}}(\alpha^k\eta^k), (\beta^k)^{-1}\operatorname{grad} f(x^{k+1}) - \mathcal{T}_{S_{\alpha^k\eta^k}}(\operatorname{grad} f(x^k))\right) \\ &= g\left((\beta^k)^{-1}\mathcal{T}_{S_{\alpha^k\eta^k}}(\alpha^k\eta^k), \operatorname{grad} f(x^{k+1})\right) - g\left(\mathcal{T}_{S_{\alpha^k\eta^k}}(\alpha^k\eta^k), \mathcal{T}_{S_{\alpha^k\eta^k}}(\operatorname{grad} f(x^k))\right) \\ &= g\left(\mathcal{T}_{R_{\alpha^k\eta^k}}(\alpha^k\eta^k), \operatorname{grad} f(x^{k+1})\right) - g\left(\alpha^k\eta^k, \operatorname{grad} f(x^k)\right) \\ &= \alpha^k\|\eta^k\|\left(g(\mathcal{T}_{R_{\alpha^k\eta^k}}(\frac{\eta^k}{\|\eta^k\|}), \operatorname{grad} f(x^{k+1})) - g\left(\frac{\eta^k}{\|\eta^k\|}, \operatorname{grad} f(x^k)\right)\right) \\ &= \alpha^k\|\eta^k\|\left(\frac{d\widetilde{m_k}(t)}{dt}\Big|_{t=\alpha^k\|\eta^k\|} - \frac{d\widetilde{m_k}(t)}{dt}\Big|_{t=0}\right) \end{aligned} \tag{3.3}$$

by Wolfe conditions, we have

$$\frac{d\widetilde{m_k}(t)}{dt}\Big|_{t=\alpha^k\|\eta^k\|} - \frac{d\widetilde{m_k}(t)}{dt}\Big|_{t=0} \geq (c_2 - 1)\frac{d\widetilde{m_k}(t)}{dt}\Big|_{t=0} = (c_2 - 1)g\left(\operatorname{grad} f(x^k), \frac{\eta^k}{\|\eta^k\|}\right) \tag{3.4}$$

then combining (3.3) and (3.4), we have

$$g(s^k, y^k) \geq (c_2 - 1)\alpha^k g(\operatorname{grad} f(x^k), \eta^k) \tag{3.5}$$

We previously prove that $\mathcal{B}^k$ is self-adjoint and we derive other results by induction. When $k = 0$, $\mathcal{B}^0$ is self-adjoint and positive definite by assumption, and $\eta^0 \neq 0$ since $\operatorname{grad} f(x^0) \neq 0$. Assume the results hold for all $i \leq k$, we then have

$$g(s^k, y^k) \geq (c_2 - 1)\alpha^k g(\operatorname{grad} f(x^k), (\mathcal{B}^k)^{-1}\operatorname{grad} f(x^k)) > 0$$

then

$$\begin{aligned}
\lambda(\mu^k) = \det\left(\mathcal{B}^{k+1}\right) &= \det\left(\tilde{\mathcal{B}}^k\right)\frac{g(y^k, s^k)}{g(s^k, \tilde{\mathcal{B}}^k s^k)}\left(1 + \phi_k\left(\frac{g(s^k, \tilde{\mathcal{B}}^k s^k)g(y^k, (\tilde{\mathcal{B}}^k)^{-1}y^k)}{g(y^k, s^k)^2} - 1\right)\right) \\
&= \det\left(\tilde{\mathcal{B}}^k\right)\frac{g(y^k, s^k)}{g(s^k, \tilde{\mathcal{B}}^k s^k)}\left(1 + \phi_k\left(\mu^k - 1\right)\right)
\end{aligned}$$

then $\lambda(\phi^k) = 0$ if and only if $\phi^k = \phi_k^c$, since $\phi^k > \phi_k^c$, we have $\lambda(\phi^k) > 0$, which means $\mathcal{B}^{k+1}$ is positive definite. Since the eigenvalue of $\mathcal{B}^{k+1}$ is continuous with respect to ϕ^k, by the intermediate value theorem, we only have to show that $\mathcal{B}^{k+1}$ is positive definite when $\phi^k = 0$, that is, the BFGS update is positive definite. We have

$$\begin{aligned}
g(\xi, \mathcal{B}^{k+1}\xi) &= g\left(\xi, \frac{g(y^k, \xi)y^k}{g(y^k, s^k)} - \frac{g(\tilde{\mathcal{B}}^k s^k, \xi)\tilde{\mathcal{B}}^k s^k}{g(s^k, \tilde{\mathcal{B}}^k s^k)}\right) + g(\xi, \tilde{\mathcal{B}}^k\xi) \\
&= \frac{g^2(y^k, \xi)}{g(y^k, s^k)} - \frac{g^2(\tilde{\mathcal{B}}^k s^k, \xi)}{g(s^k, \tilde{\mathcal{B}}^k s^k)} + g(\xi, \tilde{\mathcal{B}}^k\xi) \\
&= \frac{g^2(y^k, \xi)}{g(y^k, s^k)} + \frac{1}{g(s^k, \tilde{\mathcal{B}}^k s^k)}\left(g(s^k, \tilde{\mathcal{B}}^k s^k)g(\xi, \tilde{\mathcal{B}}^k\xi) - g^2(\tilde{\mathcal{B}}^k s^k, \xi)\right) \\
&\geq \frac{g^2(y^k, \xi)}{g(y^k, s^k)} > 0
\end{aligned}$$

then $\mathcal{B}^{k+1}$ is positive definite, the last inequality is from the Cauchy-Schwarz inequality. □

3.2 Basic Assumptions

We denote Ω_{x_0} as the level set of f at x_0, that is

$$\Omega_{x_0} = \{x \in M : f(x) \leq f(x_0)\}$$

and x^* is a local minimum of f in Ω_{x_0}. The exisitence of x^* is guaranteed if Ω_{x_0} is compact.

Definition 3.1. Let (M, g) be a Riemannian Manifold with retraction R, denote

$$\widetilde{m}_{p,\eta}(t) = f\left(R_p(t\eta)\right)$$

for all $p \in M$ and $\eta \in T_pM$ with $\|\eta\|_p = 1$. We say f is *retraction-convex* with respect to R in a subset S of M if for all $p \in S$ and $\eta \in T_pM$, the function $\widetilde{m}_{p,\eta}(t)$ is convex in t and $R_p(t\eta) \in S$ for all t. If we replace the convexity of $\widetilde{m}_{p,\eta}(t)$ with strong convexity, we say f is *retraction strongly-convex* with respect to R in S.

Assumption 3.1. *The function f is twice continuously differentiable, that is its pullback defined by $f \circ \varphi^{-1}$ is twice continuously differentiable for any chart φ.*

Assumption 3.2. *There exist $r > 0$ and $\rho > 0$ such that for each y in the restraction neighborhood defined by*

$$\widetilde{\Omega}_p := \{R_p(t\eta) : p \in \Omega_{x_0}, \|\eta\|_p = 1, t \in [0, r)\}$$

we have $\widetilde{\Omega} \subset R_y(B(0,\rho))$ where $B(0,\rho)$ is the ball in T_yM with radius ρ centered at 0 and R_y is diffeomorphism on $B(0,\rho)$.

Assumption 3.3. *For all $t \in [0, \alpha^k]$, we have $R_{x^k}(t\eta^k) \in \widetilde{\Omega}_{x^k}$. And f is strongly retraction-convex with respect to R in $\widetilde{\Omega}_{x_0}$.*

Lemma 3.3. *If Assumption 3.1 and 3.2 holds, then f is retraction-convex with respect to exponential map in a open set S if and only if $\operatorname{Hess} f$ is positive definite in S. The Hessian here is with respect to the Levi-Civita connection.*

Proof. It is easy to check it by

$$\begin{aligned}
\frac{d^2}{dt^2}\widetilde{m}_{x,\eta}(t) &= \frac{d}{dt} g_{R_x(t\eta)}\left(\operatorname{grad} f\left(R_x(t\eta)\right), \mathcal{T}^R_{t\eta}\left(\frac{\eta}{\|\eta\|}\right)\right) \\
&= g_{R_x(t\eta)}\left(\nabla_{\mathcal{T}^R_{t\eta}\left(\frac{\eta}{\|\eta\|}\right)}\operatorname{grad} f\left(R_x(t\eta)\right), \mathcal{T}^R_{t\eta}\left(\frac{\eta}{\|\eta\|}\right)\right) \\
&\quad + g_{R_x(t\eta)}\left(\operatorname{grad} f\left(R_x(t\eta)\right), \nabla_{\mathcal{T}^R_{t\eta}\left(\frac{\eta}{\|\eta\|}\right)}\mathcal{T}^R_{t\eta}\left(\frac{\eta}{\|\eta\|}\right)\right) \\
&= g_{R_x(t\eta)}\left(\operatorname{Hess} f\left(R_x(t\eta)\right)\left[\mathcal{T}^R_{t\eta}\left(\frac{\eta}{\|\eta\|}\right)\right], \mathcal{T}^R_{t\eta}\left(\frac{\eta}{\|\eta\|}\right)\right)
\end{aligned}$$

the term $g_{R_x(t\eta)}\left(\operatorname{grad} f\left(R_x(t\eta)\right), \nabla_{\mathcal{T}^R_{t\eta}\left(\frac{\eta}{\|\eta\|}\right)}\mathcal{T}^R_{t\eta}\left(\frac{\eta}{\|\eta\|}\right)\right)$ vanishes since $\mathcal{T}^R$ is exponential map, then the curve is geodesic and the covariant derivative of $\mathcal{T}^R$ along the curve is zero. $\square$

Similarly, we have the same criterion for strong retraction-convexity.

Lemma 3.4. *If Assumption 3.1 and 3.2 holds, then f is strongly retraction-convex with respect to exponential map in a open set S if and only if $\operatorname{Hess} f - \lambda_0 \mathrm{id}$ is positive definite in S for some $\lambda_0 > 0$. The Hessian here is with respect to the Levi-Civita connection.*

Lemma 3.5. *Suppose Assumption 3.1 holds, and* $\operatorname{Hess} f(x^*)$ *is positive definite. Recall denifiton of* $\widetilde{m}_{p,\eta}(t)$*, that is*

$$\widetilde{m}_{p,\eta}(t) = f\left(R_p(t\eta)\right), \quad \|\eta\|_p = 1$$

then there exist a neiborhood U *of* x^* *and two constants* $0 < a_0 < a_1$ *such that*

$$a_0 \le \frac{d^2 \widetilde{m}_{x,\eta}}{dt^2}(t) \le a_1$$

for all $x \in U$ *and* t *satisfying* $R_x(t\eta) \in U$*.*

Proof. Assume $\varphi : U \to \mathbb{R}^n$ is a chart map, and fix a moving frame $\{e_1(x), \cdots, e_n(x)\}$ on U, then for any vector field ξ defined on U, we can write

$$\xi_p = \xi^i(p) e_i(p)$$

here we use Einstein summation convention. Then we define

$$u(\hat{x}, \hat{\xi}) = f\left(R_{\varphi^{-1}(\hat{x})}\left(\xi^i e_i\left(\varphi^{-1}(\hat{x})\right)\right)\right)$$

where $\hat{x} = \varphi(x)$ is the coordinate of x in the chart, and $\hat{\xi} = (\xi^1, \cdots, \xi^n)$ is the coordinate of ξ in the moving frame. Since $\{e_i\}_{1 \le i \le n}$ is a moving frame, its Riemann metric in x is given by $G_x = I_n$.Notice that $\widetilde{m}_{x,\eta}(t) = u(\hat{x}, t\hat{\eta})$, then

$$\begin{aligned} \frac{d^2 \widetilde{m}_{x,\eta}}{dt^2}(t) &= \frac{d^2}{dt^2} u(\hat{x}, t\hat{\eta}) \\ &= \hat{\eta}^\top \operatorname{Hess}_\xi u(\hat{x}, t\hat{\eta}) \hat{\eta} \end{aligned}$$

we have $\|\hat{\eta}\|_2 = g_x(\eta, \eta) = 1$. Since f is twice continuously differentiable, then $\operatorname{Hess}_\xi u$ is continuous, and $\operatorname{Hess}_\xi u(\hat{x}^*, 0) = \operatorname{Hess} f(x^*)$ is positive definite. Therefore, there exist a neighborhood U of x^* and a constant $a_0 > 0$ such that $\operatorname{Hess}_\xi u(\hat{x}, t\hat{\eta}) \ge a_0 I_n$ for all $x \in U$ and t satisfying $R_x(t\eta) \in U$.

When this U is relatively compact, we can also get an upper bound for the eigenvalue of $\operatorname{Hess}_\xi u(\hat{x}, t\hat{\eta})$ by the continuity of $\operatorname{Hess}_\xi u$. Therefore, there exist a constant $a_1 > 0$ such that $\operatorname{Hess}_\xi u(\hat{x}, t\hat{\eta}) \le a_1 I_n$ for all $x \in U$ and t satisfying $R_x(t\eta) \in U$. □

3.3 Preliminary Results

Lemma 3.6. *If Assumption 3.1, 3.2 and 3.3 holds, then*

$$\frac{1}{2} a_0 \|s^k\|^2 \le (c_1 - 1)\alpha^k g(\operatorname{grad} f(x^k), \eta^k) \tag{3.6}$$

Proof. This proof is quite direct, we have

$$\begin{aligned} f(x^{k+1}) - f(x^k) &= \widetilde{m}_{x^k,\eta^k}(\alpha^k\|\eta^k\|) - \widetilde{m}_{x^k,\eta^k}(0) \\ &= \frac{d}{dt}\widetilde{m}_{x^k,\eta^k}(t)\Big|_{t=0}\alpha^k\|\eta^k\| + \frac{1}{2}\frac{d^2}{dt^2}\widetilde{m}_{x^k,\eta^k}(t)\Big|_{t=\tau_k}(\alpha^k\|\eta^k\|)^2 \\ &= g(\operatorname{grad} f(x^k), \alpha^k\eta^k) + \frac{1}{2}\frac{d^2}{dt^2}\widetilde{m}_{x^k,\eta^k}(t)\Big|_{t=\tau_k}(\alpha^k\|\eta^k\|)^2 \\ &\geq g(\operatorname{grad} f(x^k), \alpha^k\eta^k) + \frac{1}{2}a_0(\alpha^k\|\eta^k\|)^2 \end{aligned}$$

by Wolfe line search, we have

$$f(x^{k+1}) - f(x^k) \leq c_1\alpha^k g(\operatorname{grad} f(x^k), \eta^k)$$

combining the above two inequalities, and let $\|s^k\| = \|\mathcal{T}_{S_{\alpha^k\eta^k}}(\alpha^k\eta^k)\| = \|\alpha^k\eta^k\|$, we have

$$\frac{1}{2}a_0\|s^k\|^2 \leq (c_1 - 1)\alpha^k g(\operatorname{grad} f(x^k), \eta^k)$$

□

Lemma 3.7. *If Assumption 3.1, 3.2 and 3.3 holds, then there exist two constants* $0 < b_0 < b_1$ *such that*

$$b_0 g(s^k, s^k) \leq g(s^k, y^k) \leq b_1 g(s^k, s^k) \tag{3.7}$$

Proof. Recall that

$$s_k = \mathcal{T}_S(\alpha_k\eta_k), \quad y_k = \beta_k^{-1}\operatorname{grad} f(x_{k+1}) - \mathcal{T}_S(\operatorname{grad} f(x_k))$$

then

$$\begin{aligned} g(s^k, y^k) &= g\left(\mathcal{T}_S(\alpha_k\eta_k), \beta_k^{-1}\operatorname{grad} f(x_{k+1}) - \mathcal{T}_S(\operatorname{grad} f(x_k))\right) \\ &= \beta_k^{-1} g\left(\mathcal{T}_S(\alpha_k\eta_k),\ \operatorname{grad} f(x_{k+1})\right) - g(\mathcal{T}_S(\alpha_k\eta_k),\ \mathcal{T}_S(\operatorname{grad} f(x_k))) \end{aligned}$$

since $\mathcal{T}_S$ is isometric, and β_k is defined by

$$\beta_k = \frac{\|\alpha_k\eta_k\|}{\|\mathcal{T}_R(\alpha_k\eta_k)\|} = \frac{\|s^k\|}{\|\mathcal{T}_R(\alpha_k\eta_k)\|}$$

by locking condition, we have $\beta_k^{-1} g\left(\mathcal{T}_S(\alpha_k\eta_k),\, \mathrm{grad}\, f(x_{k+1})\right) = g\left(\mathcal{T}_R(\alpha_k\eta_k),\, \mathrm{grad}\, f(x_{k+1})\right)$, then

$$
\begin{aligned}
g(s^k, y^k) &= g\left(\mathcal{T}_R(\alpha_k\eta_k),\, \mathrm{grad}\, f(x_{k+1})\right) - g(\mathcal{T}_S(\alpha_k\eta_k),\, \mathcal{T}_S(\mathrm{grad}\, f(x_k))) \\
&= g\left(\mathcal{T}_R(\alpha_k\eta_k),\, \mathrm{grad}\, f(x_{k+1})\right) - g(\alpha_k\eta_k,\, \mathrm{grad}\, f(x_k)) \\
&= \alpha_k\|\eta_k\| \left(g\left(\mathcal{T}_R\left(\frac{\eta_k}{\|\eta_k\|}\right),\, \mathrm{grad}\, f(x_{k+1})\right) - g\left(\frac{\eta_k}{\|\eta_k\|},\, \mathrm{grad}\, f(x_k)\right)\right) \\
&= \alpha_k\|\eta_k\| \left(\frac{d}{dt} f\left(R_{x^k}(t\eta^k/\|\eta^k\|)\right)\Big|_{t=\alpha^k\|\eta^k\|} - \frac{d}{dt} f\left(R_{x^k}(t\eta^k/\|\eta^k\|)\right)\Big|_{t=0}\right) \\
&= \alpha_k\|\eta_k\| \left(\frac{d}{dt}\widetilde{m}_k(t)\Big|_{t=\alpha^k\|\eta^k\|} - \frac{d}{dt}\widetilde{m}_k(t)\Big|_{t=0}\right) \\
&= \alpha_k\|\eta_k\| \int_0^{\alpha^k\|\eta^k\|} \frac{d^2}{dt^2}\widetilde{m}_k(t)dt
\end{aligned}
$$

then

$$
\frac{g(s^k, y^k)}{g(s^k, s^k)} = \frac{1}{\alpha_k\|\eta_k\|} \int_0^{\alpha^k\|\eta^k\|} \frac{d^2}{dt^2}\widetilde{m}_k(t)dt
$$

since f is strongly retraction-convex with respect to R in $\overline{\widetilde{\Omega}_{x_0}}$, we have $b_0 \le \frac{d^2}{dt^2}\widetilde{m}_k(t) \le b_1$ for some constants $0 < b_0 < b_1$, then

$$
b_0 \le \frac{g(s^k, y^k)}{g(s^k, s^k)} \le b_1
$$

□

Lemma 3.8. *If Assumption 3.1, 3.2 and 3.3 holds, then there exist two positive constants* $0 < a_2 < a_3$ *such that*

$$
a_2\|\mathrm{grad}\, f(x^k)\| \cos\theta_k \le \|s^k\| \le a_3\|\mathrm{grad}\, f(x^k)\| \cos\theta_k \tag{3.8}
$$

for all $k \ge 0$, *where* θ_k *is the angle between* $\mathrm{grad}\, f(x^k)$ *and* η^k *defined by*

$$
\cos\theta_k = \frac{-g(\mathrm{grad}\, f(x^k), \eta^k)}{\|\mathrm{grad}\, f(x^k)\|\|\eta^k\|}
$$

Proof. By (3.5), we have

$$
\begin{aligned}
g(s^k, y^k) &\ge (c_2 - 1)\alpha^k g(\mathrm{grad}\, f(x^k), \eta^k) = \|\mathrm{grad}\, f(x^k)\|\|\eta^k\| \cos\theta_k \\
&= \alpha^k(1 - c_2)\|s^k\|\|\mathrm{grad}\, f(x^k)\| \cos\theta_k
\end{aligned}
$$

then by (3.7) we obtain

$$
\|s^k\| \ge a_2\|\mathrm{grad}\, f(x^k)\| \cos\theta_k
$$

where $a_2 = (1 - c_2)/b_1$.

On the other hand, by (3.6), we have

$$(c_1 - 1)\alpha^k g(\operatorname{grad} f(x^k), \eta^k) \geq \frac{1}{2} a_0 \|s^k\|^2$$

which means

$$(1 - c_1)\|\operatorname{grad} f(x^k)\| \|s^k\| \cos\theta_k \geq \frac{1}{2} a_0 \|s^k\|^2$$

then

$$\|s^k\| \leq a_3 \|\operatorname{grad} f(x^k)\| \cos\theta_k$$

where $a_3 = 2(1 - c_1)/a_0$. □

The following lemma shows the two vector transportations $\mathcal{T}_S$ and $\mathcal{T}_R$ are close to each other when the step size is small, which is a direct consequence of the locking condition.

Lemma 3.9. *Let $\mathcal{T}_1 \in C^0$ and $\mathcal{T}_2 \in C^\infty$ where $\mathcal{T}_1$ and $\mathcal{T}_2$ are two vector transportations on M along R. If $\mathcal{T}_1$ and $\mathcal{T}_2$ satisfies the locking condition, then for any $\overline{x} \in M$, there exists a constant $a_4 > 0$ and a neighborhood U of $\overline{x}$ such that for all $x, y \in U$ and $\xi \in T_x M$, we have*

$$\|(\mathcal{T}_1 - \mathcal{T}_2)_\xi\| \leq a_4 \|\xi\| \|\eta\|$$

where $\eta = R_x^{-1}(y)$ is the tangent vector at x such that $R_x(\eta) = y$.

Proof. Choose an orthonormal frame $\{e_i(x)\}$ in a neighborhood of $\overline{x}$, so that the metric matrix $G_x = I$ throughout this neighborhood. For any x in this neighborhood and $\eta \in T_x M$, define the coordinate matrices $L_R(\hat{x}, \hat{\eta})$ and $L_2(\hat{x}, \hat{\eta})$ of $\mathcal{T}_{R,\eta}$ and $\mathcal{T}_{2,\eta}$ respectively by

$$\mathcal{T}_{R_\eta}\xi = \sum_i \big(L_R(\hat{x}, \hat{\eta})\hat{\xi}\big)^i e_i\big(R_x(\eta)\big), \qquad \mathcal{T}_{2_\eta}\xi = \sum_i \big(L_2(\hat{x}, \hat{\eta})\hat{\xi}\big)^i e_i\big(R_x(\eta)\big),$$

where $\hat{\xi}, \hat{\eta} \in \mathbb{R}^n$ are the coordinate representations of ξ, η in the orthonormal frame. Concretely,

$$\big(L_R(\hat{x}, \hat{\eta})\big)_{ij} = g\big(e^i(R_x(\eta)),\, dR_x\big|_\eta [e_j(x)]\big), \qquad \big(L_2(\hat{x}, \hat{\eta})\big)_{ij} = g\big(e^i(R_x(\eta)),\, \mathcal{T}_{2,\eta}[e_j(x)]\big).$$

by the triangle inequality, inserting $\mathcal{T}_{R,\eta}\xi$ as an intermediate term,

$$\|\mathcal{T}_{1_\eta}\xi - \mathcal{T}_{2_\eta}\xi\| \leq \underbrace{\|\mathcal{T}_{1_\eta}\xi - \mathcal{T}_{R_\eta}\xi\|}_{(A)} + \underbrace{\|\mathcal{T}_{R_\eta}\xi - \mathcal{T}_{2_\eta}\xi\|}_{(B)}. \tag{3.9}$$

Since $\mathcal{T}_1$ satisfies the locking condition, by definition we have $\|\mathcal{T}_{1_\eta} - \mathcal{T}_{R_\eta}\| \leq \delta\|\eta\|$, hence

$$(A) = \|(\mathcal{T}_{1_\eta} - \mathcal{T}_{R_\eta})\xi\| \leq \delta\|\eta\|\|\xi\| =: b_0\|\xi\|\|\eta\|.$$

In the orthonormal moving frame, we have

$$(B) \leq b_1 \|(L_R(\hat{x},\hat{\eta}) - L_2(\hat{x},\hat{\eta}))\hat{\xi}\|_2 \leq b_1 \|\hat{\xi}\|_2 \, \|L_R(\hat{x},\hat{\eta}) - L_2(\hat{x},\hat{\eta})\|_2.$$

We notice that at $\hat{\eta} = 0$, both matrix-valued functions reduce to the identity. Hence $L_R(\hat{x},0) - L_2(\hat{x},0) = 0$. Since $\mathcal{T}_2 \in C^\infty$, the map $\hat{\eta} \mapsto L_2(\hat{x},\hat{\eta})$ is smooth, and a first-order Taylor expansion in $\hat{\eta}$ around 0 gives

$$\|L_R(\hat{x},\hat{\eta}) - L_2(\hat{x},\hat{\eta})\|_2 \leq b_2' \|\hat{\eta}\|_2,$$

so $(B) \leq b_2 \|\xi\| \|\eta\|$ for some constant $b_2 > 0$.

Substituting both estimates into (3.9),

$$\|\mathcal{T}_{1_\eta}\xi - \mathcal{T}_{2_\eta}\xi\| \leq (b_0 + b_2)\|\xi\|\|\eta\| =: a_4 \|\xi\|\|\eta\|. \qquad \square$$

The following lemma is a direct consequence of the previous lemma, which shows that $\mathcal{T}_S$ and $\mathcal{T}_R$ are close to each other when the step size is small.

Lemma 3.10. *Let M is a Riemannian manifold with retraction R and $\mathcal{T}_R$ denotes the differentiated retraction of R. Then there is a neighborhood U of $\overline{x}$ and a constant $\tilde{a}_4 > 0$ such that for all $x \in U$, $\eta = R_x^{-1} y$ and $\xi \in T_x M$ with $\|\xi\| = 1$, we have*

$$\left| \|\mathcal{T}_{R_\eta}\xi\| - 1 \right| \leq \tilde{a}_4 \|\eta\|$$

Proof. In Lemma 3.8 we choose $\mathcal{T}_1 = \mathcal{T}_S$ and $\mathcal{T}_2 = \mathcal{T}_R$ then the proof is completed by the previous lemma. $\square$

The following lemma is a generalization of the Zoutendijk condition in the Euclidean case.

Lemma 3.11. *If Assumption 3.1, 3.2 and 3.3 holds, then for all $k \geq 0$, there exists a constant $a_5 > 0$ such that*

$$f(x^{k+1}) - f(x^k) \leq (1 - a_5 \cos^2\theta_k) f(x^k) - f(x^*)$$

where θ_k is the angle between $\operatorname{grad} f(x^k)$ *and η^k defined by*

$$\cos\theta_k = \frac{-g(\operatorname{grad} f(x^k), \eta^k)}{\|\operatorname{grad} f(x^k)\|\|\eta^k\|}$$

Proof. Let $z^k = \|R_{x^*}^{-1}(x^k)\|$, and ζ^k is its unit vector, that is $\zeta^k = R_{x^*}^{-1}(x^k)/z^k$. We define

$$m_k(t) = f\left(R_{x^*}(z^k + t\zeta^k)\right)$$

then by Taylor expansion, we have

$$m_k(0) - m_k(z^k) = -\frac{d}{dt}m_k(t)\Big|_{t=0} z^k - \frac{1}{2}\frac{d^2}{dt^2}m_k(t)\Big|_{t=\tau_k}(z^k)^2 \tag{3.10}$$

where $\tau_k \in [0, z^k]$. By Assumption 3.3, we have

$$\frac{d}{dt}m_k(t)\Big|_{t=z^k} \geq \frac{1}{2}a_0 z^k$$

since the second derivative of m_k is bounded from below by a_0, then

$$\begin{aligned} f(x^k) - f(x^*) &\leq \frac{d}{dt}m_k(t)\Big|_{t=0} z^k \\ &\leq \frac{2}{a_0}\left(\frac{d}{dt}m_k(t)\Big|_{t=0}\right)^2 \\ &= \frac{2}{a_0}g^2\left(\operatorname{grad} f(x^k), \mathcal{T}_{R_{z^k\zeta^k}}(\zeta^k)\right) \\ &\leq b_0\|\operatorname{grad} f(x^k)\|^2 \end{aligned} \tag{3.11}$$

where b_0 is a positive constant. On the other hand, by Lemma 3.6, we have

$$\alpha^k g(\operatorname{grad} f(x^k), \eta^k) \leq -\frac{a_0}{2(1-c_1)}\|s^k\|^2$$

Substituting into the first Wolfe condition (2.18), we obtain

$$f(x^{k+1}) - f(x^k) \leq c_1\alpha^k g(\operatorname{grad} f(x^k), \eta^k) \leq -\frac{c_1 a_0}{2(1-c_1)}\|s^k\|^2$$

By Lemma 3.8, $\|s^k\| \geq a_2\|\operatorname{grad} f(x^k)\|\cos\theta_k$, hence

$$f(x^{k+1}) - f(x^k) \leq -\frac{c_1 a_0 a_2^2}{2(1-c_1)}\|\operatorname{grad} f(x^k)\|^2\cos^2\theta_k =: -b_1\|\operatorname{grad} f(x^k)\|^2\cos^2\theta_k$$

where $b_1 = \frac{c_1 a_0 a_2^2}{2(1-c_1)} > 0$. Combining with the previous estimate

$$f(x^k) - f(x^*) \leq b_0\|\operatorname{grad} f(x^k)\|^2$$

we get

$$\begin{aligned} f(x^{k+1}) - f(x^*) &= \left(f(x^{k+1}) - f(x^k)\right) + \left(f(x^k) - f(x^*)\right) \\ &\leq -b_1\|\operatorname{grad} f(x^k)\|^2\cos^2\theta_k + \left(f(x^k) - f(x^*)\right) \\ &\leq -\frac{b_1}{b_0}\cos^2\theta_k\left(f(x^k) - f(x^*)\right) + \left(f(x^k) - f(x^*)\right) \\ &= \left(1 - a_5\cos^2\theta_k\right)\left(f(x^k) - f(x^*)\right) \end{aligned}$$

where $a_5 = b_1/b_0 > 0$. $\square$

Lemma 3.12. *If Assumption 3.1, 3.2 and 3.3 holds, then there exists two constants* $0 < a_6 < a_7$ *such that*

$$a_6 \frac{g(s^k, \tilde{B}^k s^k)}{\|s^k\|^2} \leq \alpha^k \leq a_7 \frac{g(s^k, \tilde{B}^k s^k)}{\|s^k\|^2} \tag{3.12}$$

Proof. By Lemma 3.6 and 3.2, we have

$$\begin{aligned}(1-c_2) g\left(s^k, \tilde{B}^k s^k\right) &= (c_2 - 1)(\alpha^k)^2 g\left(\operatorname{grad} f(x^k), \eta^k\right) \\ &\leq \alpha^k g(s^k, y^k) \leq \alpha^k a_1 \|s^k\|^2\end{aligned}$$

therefore,

$$\alpha^k \geq a_6 \frac{g(s^k, \tilde{B}^k s^k)}{\|s^k\|^2}$$

where $a_6 = (1-c_2)/a_1 > 0$.

On the other hand, by Lemma 3.6 and 3.2, we have

$$(c_1 - 1)\alpha^k g\left(\operatorname{grad} f(x^k), \eta^k\right) = (1-c_1) g\left(s^k, \tilde{B}^k s^k\right) = \frac{1-c_1}{\alpha^k} g\left(s^k, \tilde{B}^k s^k\right) \geq \frac{a_0}{2}\|s^k\|^2$$

then

$$\alpha^k \leq a_7 \frac{g(s^k, \tilde{B}^k s^k)}{\|s^k\|^2}$$

where $a_7 = 2(1-c_1)/a_0 > 0$. □

Lemma 3.13. *Suppose Assumption 3.1 holds, then for all* $k \geq 0$*, there is a constant* $a_8 > 0$ *such that*

$$\|y^k\|^2 \leq a_8 g(s^k, y^k) \tag{3.13}$$

Proof. We define

$$y_P^k = \operatorname{grad} f(x_{k+1}) - P_{0\to 1}^{\gamma^k}(\operatorname{grad} f(x_k))$$

where P is the parallel transportation along the retraction curve γ^k connecting x^k and x^{k+1} such that $\gamma^k(t) = R_{x^k}(t\alpha^k\eta^k)$ for $t \in [0,1]$. We first show that

$$\|P_{1\to 0}^{\gamma^k} y_P^k - \overline{H}_k \alpha^k \eta^k\| \leq b_0 \|s^k\|^2 \tag{3.14}$$

where $\overline{H}_k$ is defined by

$$\overline{H}_k = \int_0^1 P_{t\to 0}^{\gamma^k} \operatorname{Hess} f(\gamma^k(t)) P_{0\to t}^{\gamma^k} dt \tag{3.15}$$

we define curve $\phi(t)$ by

$$\phi(t) = P_{t\to 0}^{\gamma^k} \operatorname{grad} f(\gamma^k(t))$$

then $\phi(0) = \operatorname{grad} f(x^k)$ and $\phi(1) = P^{\gamma^k}_{1\to 0}\operatorname{grad} f(x_{k+1})$. Then

$$P^{\gamma^k}_{1\to 0} y^k_P = \phi(1) - \phi(0) = \int_0^1 \phi'(t)dt$$

we notice that

$$\phi'(t) = P^{\gamma^k}_{t\to 0} \nabla_{\dot\gamma^k(t)} \operatorname{grad} f(\gamma^k(t)) = P^{\gamma^k}_{t\to 0} \operatorname{Hess} f(\gamma^k(t))[\dot\gamma^k(t)]$$

then

$$P^{\gamma^k}_{1\to 0} y^k_P = \int_0^1 P^{\gamma^k}_{t\to 0} \operatorname{Hess} f(\gamma^k(t))[\dot\gamma^k(t)]\, dt$$

now we have

$$P^{\gamma^k}_{1\to 0} y^k_P - \overline{H}_k \alpha^k \eta^k = \int_0^1 P^{\gamma^k}_{t\to 0} \operatorname{Hess} f(\gamma^k(t)) \Big[\dot\gamma^k(t) - P^{\gamma^k}_{0\to t}(\alpha^k\eta^k)\Big]\, dt$$

It remains to estimate $\dot\gamma^k(t) - P^{\gamma^k}_{0\to t}(\alpha^k\eta^k)$. Let $V(t) = P^{\gamma^k}_{0\to t}(\alpha^k\eta^k)$ be the parallel transport of the initial velocity along γ^k, and define $W(t) = \dot\gamma^k(t) - V(t)$. Since R is a retraction, by the first-order condition $dR_{x^k}\big|_0 = \mathrm{id}$, we have

$$W(0) = \dot\gamma^k(0) - \alpha^k\eta^k = dR_{x^k}\big|_0[\alpha^k\eta^k] - \alpha^k\eta^k = 0$$

Since V is a parallel field ($\nabla_{\dot\gamma^k} V = 0$), we have

$$\nabla_{\dot\gamma^k} W = \nabla_{\dot\gamma^k} \dot\gamma^k$$

which is the "acceleration" of the retraction curve. Define $\hat W(t) = P^{\gamma^k}_{t\to 0} W(t)$, which pulls $W(t)$ back to $T_{x^k}M$ via parallel transport. Since parallel transport commutes with covariant differentiation along γ^k, we have

$$\hat W'(t) = P^{\gamma^k}_{t\to 0} \left(\nabla_{\dot\gamma^k} W\right)(t) = P^{\gamma^k}_{t\to 0} \left(\nabla_{\dot\gamma^k} \dot\gamma^k\right)(t)$$

Since $\gamma^k(t) = R_{x^k}(t\alpha^k\eta^k)$ and R is smooth, the acceleration $\nabla_{\dot\gamma^k}\dot\gamma^k$ is bounded on any compact set. More precisely, there is a constant $C > 0$ (depending only on R and the compact neighborhood) such that

$$\|\nabla_{\dot\gamma^k}\dot\gamma^k(t)\| \le C\|\alpha^k\eta^k\|^2 = C\|s^k\|^2$$

for all $t \in [0,1]$. Since parallel transport is an isometry, $\|\hat W'(t)\| = \|\nabla_{\dot\gamma^k}\dot\gamma^k(t)\| \le C\|s^k\|^2$. Integrating from 0 to t with $\hat W(0) = W(0) = 0$, we obtain

$$\|W(t)\| = \|\hat W(t)\| = \left\| \int_0^t \hat W'(\tau)\, d\tau \right\| \le Ct\|s^k\|^2$$

for all $t \in [0,1]$. Since parallel transport is an isometry and $\operatorname{Hess} f$ is uniformly bounded (say by $L > 0$) on the compact set $\overline{\widetilde{\Omega}_{x_0}}$, we conclude

$$\left\| P_{1\to 0}^{\gamma^k} y_P^k - \overline{H}_k \alpha^k \eta^k \right\| \leq \int_0^1 L \cdot Ct \|s^k\|^2 \, dt = \frac{CL}{2} \|s^k\|^2 =: b_0 \|s^k\|^2$$

Now we have

$$\begin{aligned} \|y^k\| \leq \|y_P^k\| + \|y^k - y_P^k\| &\leq \|P_{1\to 0}^{\gamma^k} y_P^k\| + \|P_{1\to 0}^{\gamma^k} y_P^k - \overline{H}_k \alpha^k \eta^k\| + \|\overline{H}_k \alpha^k \eta^k - y^k\| \\ &\leq \|P_{1\to 0}^{\gamma^k} y_P^k\| + b_0 \|s^k\|^2 + \|\overline{H}_k \alpha^k \eta^k - y^k\| \\ &\leq \|P_{1\to 0}^{\gamma^k} y_P^k\| + b_0 \|s^k\|^2 + \|\overline{H}_k - \tilde{B}^k\| \|s^k\| \\ &\leq \|P_{1\to 0}^{\gamma^k} y_P^k\| + b_1 \|s^k\|^2 \end{aligned}$$

where $b_1 = b_0 + \sup_k \|\overline{H}_k - \tilde{B}^k\|$. Since $\tilde{B}^k$ is uniformly positive definite, there is a constant $a_8 > 0$ such that $\|P_{1\to 0}^{\gamma^k} y_P^k\|^2 \leq a_8 g(s^k, y^k)$, hence

$$\|y^k\|^2 \leq a_8 g(s^k, y^k) + b_2 \|s^k\|^4$$

where $b_2 = b_1^2 + 2\sqrt{a_8} b_1$. Since $\|s^k\|$ is bounded, we can choose a constant $a_9 > 0$ such that $b_2 \|s^k\|^4 \leq a_9 g(s^k, y^k)$ for all k, hence

$$\|y^k\|^2 \leq (a_8 + a_9) g(s^k, y^k) =: a_8' g(s^k, y^k). \qquad \square$$

replacing a_8 by a_8' completes the proof.

Lemma 3.14. *Suppose Assumptions 3.1, 3.2 and 3.3 hold. Then there exist constants $a_{10} > 0$, $a_{11} > 0$, $a_{12} > 0$ such that*

$$\frac{g(s^k, \tilde{\mathcal{B}}^k s^k)}{g(s^k, y^k)} \leq a_{10} \alpha^k \tag{3.16}$$

$$\frac{\|\tilde{\mathcal{B}}^k s^k\|^2}{g(s^k, \tilde{\mathcal{B}}^k s^k)} \geq a_{11} \frac{\alpha^k}{\cos^2 \theta_k} \tag{3.17}$$

$$\frac{|g(y^k, \tilde{\mathcal{B}}^k s^k)|}{g(y^k, s^k)} \leq a_{12} \frac{\alpha^k}{\cos \theta_k} \tag{3.18}$$

for all k.

Proof. Proof of (3.16)**.** By (3.5), we have

$$g(s^k, y^k) \geq (c_2 - 1) \alpha^k g(\operatorname{grad} f(x^k), \eta^k)$$

Since $\mathcal{B}^k\eta^k = -\mathrm{grad}\, f(x^k)$ (Step 3 of Algorithm 2.1) and $\tilde{\mathcal{B}}^k = \mathcal{T}_S \circ \mathcal{B}^k \circ \mathcal{T}_S^{-1}$, we have $\tilde{\mathcal{B}}^k s^k = -\alpha^k \mathcal{T}_S(\mathrm{grad}\, f(x^k))$. Since $\mathcal{T}_S$ is isometric,

$$g(s^k, \tilde{\mathcal{B}}^k s^k) = -\alpha^k g(\mathcal{T}_S(\alpha^k\eta^k), \mathcal{T}_S(\mathrm{grad}\, f(x^k))) = -(\alpha^k)^2 g(\eta^k, \mathrm{grad}\, f(x^k))$$

Therefore,

$$g(s^k, y^k) \geq (1-c_2)\alpha^k \left(-g(\mathrm{grad}\, f(x^k), \eta^k)\right) = \frac{1-c_2}{\alpha^k} g(s^k, \tilde{\mathcal{B}}^k s^k)$$

and thus

$$\frac{g(s^k, \tilde{\mathcal{B}}^k s^k)}{g(s^k, y^k)} \leq a_{10}\alpha^k$$

where $a_{10} = 1/(1-c_2)$.

Proof of (3.17)**.** By Step 3 of Algorithm 2.1 and the definition of $\cos\theta_k$,

$$\frac{\|\tilde{\mathcal{B}}^k s^k\|^2}{g(s^k, \tilde{\mathcal{B}}^k s^k)} = \frac{(\alpha^k)^2 \|\mathrm{grad}\, f(x^k)\|^2}{(\alpha^k)^2 \|\eta^k\| \|\mathrm{grad}\, f(x^k)\| \cos\theta_k} = \frac{\alpha^k \|\mathrm{grad}\, f(x^k)\|}{\|s^k\| \cos\theta_k}$$

By Lemma 3.8, $\|s^k\| \leq a_3 \|\mathrm{grad}\, f(x^k)\| \cos\theta_k$, hence

$$\frac{\|\tilde{\mathcal{B}}^k s^k\|^2}{g(s^k, \tilde{\mathcal{B}}^k s^k)} \geq \frac{\alpha^k \|\mathrm{grad}\, f(x^k)\|}{a_3 \|\mathrm{grad}\, f(x^k)\| \cos^2\theta_k} = \frac{\alpha^k}{a_3 \cos^2\theta_k} =: a_{11} \frac{\alpha^k}{\cos^2\theta_k}$$

where $a_{11} = 1/a_3 > 0$.

Proof of (3.18)**.** By Step 3 of Algorithm 2.1,

$$\frac{|g(y^k, \tilde{\mathcal{B}}^k s^k)|}{g(s^k, y^k)} \leq \frac{\alpha^k \|y^k\| \|\mathrm{grad}\, f(x^k)\|}{g(s^k, y^k)}$$

By Lemma 3.7, $\|y^k\|^2 \leq a_8 g(s^k, y^k)$, so $\|y^k\| \leq a_8^{1/2} g^{1/2}(s^k, y^k)$. Substituting,

$$\frac{|g(y^k, \tilde{\mathcal{B}}^k s^k)|}{g(s^k, y^k)} \leq \frac{a_8^{1/2} \alpha^k \|\mathrm{grad}\, f(x^k)\|}{g^{1/2}(s^k, y^k)}$$

By (3.7), $g(s^k, y^k) \geq b_0 \|s^k\|^2$, so $g^{1/2}(s^k, y^k) \geq b_0^{1/2} \|s^k\|$. Then

$$\frac{|g(y^k, \tilde{\mathcal{B}}^k s^k)|}{g(s^k, y^k)} \leq \frac{a_8^{1/2} \alpha^k \|\mathrm{grad}\, f(x^k)\|}{b_0^{1/2} \|s^k\|}$$

By Lemma 3.8, $\|s^k\| \geq a_2 \|\mathrm{grad}\, f(x^k)\| \cos\theta_k$, hence

$$\frac{|g(y^k, \tilde{\mathcal{B}}^k s^k)|}{g(s^k, y^k)} \leq \frac{a_8^{1/2}}{b_0^{1/2} a_2} \cdot \frac{\alpha^k}{\cos\theta_k} =: a_{12} \frac{\alpha^k}{\cos\theta_k}$$

where $a_{12} = a_8^{1/2}/(b_0^{1/2} a_2) > 0$. $\square$

Lemma 3.15. *Suppose Assumptions 3.1, 3.2 and 3.3 hold, and $\phi_k \in [0,1]$. Then there exists a constant $a_{13} > 0$ such that*

$$\prod_{j=1}^{k} \alpha^j \geq a_{13}^k \tag{3.19}$$

for all $k \geq 1$.

Proof. Let $\hat{B}_k$ denote the coordinate matrix of $\mathcal{B}^k$ with respect to an orthonormal frame $\{e_i\}$. Since $\mathcal{T}_S$ is an isometric vector transport, the push-out $\tilde{\mathcal{B}}^k = \mathcal{T}_S \circ \mathcal{B}^k \circ \mathcal{T}_S^{-1}$ preserves both the trace and the determinant in coordinates, that is,

$$\operatorname{trace}(\hat{B}_k) = \operatorname{trace}(\hat{\tilde{B}}_k), \qquad \det(\hat{B}_k) = \det(\hat{\tilde{B}}_k) \tag{3.20}$$

From the Broyden update formula (2.16), taking the trace in coordinates gives

$$\begin{aligned} \operatorname{trace}(\hat{B}_{k+1}) = \operatorname{trace}(\hat{B}_k) + \frac{\|y^k\|^2}{g(y^k, s^k)} \left(1 + \phi_k \frac{g(s^k, \tilde{\mathcal{B}}^k s^k)}{g(y^k, s^k)}\right) \\ + (\phi_k - 1) \frac{\|\tilde{\mathcal{B}}^k s^k\|^2}{g(s^k, \tilde{\mathcal{B}}^k s^k)} - 2\phi_k \frac{g(y^k, \tilde{\mathcal{B}}^k s^k)}{g(y^k, s^k)} \end{aligned} \tag{3.21}$$

We now bound each term. Since $\phi_k \leq 1$, the third term

$$(\phi_k - 1) \frac{\|\tilde{\mathcal{B}}^k s^k\|^2}{g(s^k, \tilde{\mathcal{B}}^k s^k)} \leq 0$$

by the estimate $\|y^k\|^2 \leq a_8\, g(s^k, y^k)$, we have

$$\frac{\|y^k\|^2}{g(y^k, s^k)} \leq a_8$$

By (3.16) that is

$$\frac{g(s^k, \tilde{\mathcal{B}}^k s^k)}{g(y^k, s^k)} \leq a_{10} \alpha^k$$

and (3.18), that is

$$\frac{|g(y^k, \tilde{\mathcal{B}}^k s^k)|}{g(y^k, s^k)} \leq a_{12} \frac{\alpha^k}{\cos\theta_k}$$

substituting these into (3.21)

$$\operatorname{trace}(\hat{B}_{k+1}) \leq \operatorname{trace}(\hat{B}_k) + a_8(1 + a_{10}\alpha^k) + 2a_{12} \frac{\alpha^k}{\cos\theta_k} \tag{3.22}$$

It remains to bound $\frac{\alpha^k}{\cos\theta_k}$. By Lemma 3.12, we have

$$\alpha^k \leq a_7 \frac{g(s^k, \tilde{\mathcal{B}}^k s^k)}{\|s^k\|^2}$$

since

$$g(s^k, \tilde{\mathcal{B}}^k s^k) = (\alpha^k)^2 \|\eta^k\| \|\operatorname{grad} f(x^k)\| \cos\theta_k, \quad \|s^k\|^2 = (\alpha^k)^2 \|\eta^k\|^2$$

we obtain

$$\frac{\alpha^k}{\cos\theta_k} \leq a_7 \frac{\|\operatorname{grad} f(x^k)\|}{\|\eta^k\|} = a_7 \frac{\|\tilde{\mathcal{B}}^k s^k\|}{\|s^k\|} \tag{3.23}$$

since $\tilde{\mathcal{B}}^k$ is positive definite, the largest eigenvalue is bounded by the trace:

$$\frac{\|\tilde{\mathcal{B}}^k s^k\|}{\|s^k\|} \leq \lambda_{\max}(\tilde{\mathcal{B}}^k) \leq \operatorname{trace}(\hat{\hat{B}}_k) = \operatorname{trace}(\hat{B}_k)$$

Similarly, $\alpha^k \leq a_7 \frac{g(s^k, \tilde{\mathcal{B}}^k s^k)}{\|s^k\|^2} \leq a_7 \operatorname{trace}(\hat{B}_k)$. Substituting into (3.22)

$$\operatorname{trace}(\hat{B}_{k+1}) \leq \operatorname{trace}(\hat{B}_k)\,(1 + a_8 a_{10} a_7 + 2a_{12} a_7) + a_8$$

Let $c_0 = 1 + a_8 a_{10} a_7 + 2a_{12} a_7 > 1$. By induction,

$$\operatorname{trace}(\hat{B}_k) \leq c_0^k \operatorname{trace}(\hat{B}_0) + \frac{a_8(c_0^k - 1)}{c_0 - 1}$$

hence there exists a constant $b_0 > 1$ such that

$$\operatorname{trace}(\hat{B}_k) \leq b_0^k \tag{3.24}$$

for all $k \geq 0$.

From the determinant formula for the Broyden update (see the proof of Lemma 3.2), since $\phi_k \in [0, 1]$ and $\mu^k \geq 1$ by Cauchy–Schwarz inequality, we have $1 + \phi_k(\mu^k - 1) \geq 1$, and therefore

$$\det(\hat{B}_{k+1}) \geq \det(\hat{\hat{B}}_k) \frac{g(y^k, s^k)}{g(s^k, \tilde{\mathcal{B}}^k s^k)} = \det(\hat{B}_k) \frac{g(y^k, s^k)}{g(s^k, \tilde{\mathcal{B}}^k s^k)} \tag{3.25}$$

By (3.16), $\frac{g(s^k, \tilde{\mathcal{B}}^k s^k)}{g(s^k, y^k)} \leq a_{10} \alpha^k$, so $\frac{g(y^k, s^k)}{g(s^k, \tilde{\mathcal{B}}^k s^k)} \geq \frac{1}{a_{10}\alpha^k}$. Iterating (3.25) then it turns out that

$$\det(\hat{B}_{k+1}) \geq \det(\hat{B}_1) \prod_{j=1}^{k} \frac{1}{a_{10}\alpha^j} \tag{3.26}$$

By the arithmetic-geometric mean inequality applied to the eigenvalues of the positive definite matrix $\hat{B}_{k+1}$ then we have

$$\det(\hat{B}_{k+1}) \leq \left(\frac{\operatorname{trace}(\hat{B}_{k+1})}{d}\right)^d$$

where $d = \dim M$. Combining with (3.24) and (3.26) we obtain

$$\det(\hat{B}_1) \prod_{j=1}^{k} \frac{1}{a_{10}\alpha^j} \leq \left(\frac{b_0^{k+1}}{d}\right)^d$$

that is,

$$\prod_{j=1}^{k} \frac{1}{\alpha^j} \leq \frac{a_{10}^k \, b_0^{d(k+1)}}{d^d \, \det(\hat{B}_1)}$$

and hence

$$\prod_{j=1}^{k} \alpha^j \geq \frac{d^d \, \det(\hat{B}_1)}{b_0^d} \cdot \frac{1}{(a_{10} \, b_0^d)^k}$$

Let $C = \frac{d^d \, \det(\hat{B}_1)}{b_0^d} > 0$ and $\tilde{a} = \frac{1}{a_{10} \, b_0^d} > 0$. Then $\prod_{j=1}^{k} \alpha^j \geq C \, \tilde{a}^k$. Since $C > 0$ is a fixed constant (independent of k) and $C^{1/k} \to 1$ as $k \to \infty$, we can choose $a_{13} = \tilde{a}/2 > 0$ (or any constant strictly smaller than $\tilde{a}$) such that $C \, \tilde{a}^k \geq a_{13}^k$ for all $k \geq 1$. This completes the proof. □

3.4 Main Result

Theorem 3.1. *Suppose Assumptions 3.1, 3.2 and 3.3 hold, and $\phi_k \in [0, 1-\delta]$ for some fixed $\delta > 0$. Then the sequence $\{x^k\}$ generated by Algorithm 2.1 converges to the minimizer x^* of f.*

Proof. The trace inequality (3.21) can be rewritten as

$$\operatorname{trace}(\hat{B}_{k+1}) \leq \operatorname{trace}(\hat{B}_k) + a_8 + t_k \alpha^k \tag{3.27}$$

where

$$t_k = \phi_k a_8 a_{10} - \frac{a_{11}(1-\phi_k)}{\cos^2 \theta_k} + \frac{2\phi_k a_{12}}{\cos \theta_k}$$

Here the term $-\frac{a_{11}(1-\phi_k)}{\cos^2 \theta_k}$ arises from the third term in (3.21) via (3.17), which was previously dropped but is now retained.

We argue by contradiction. Assume $\cos \theta_k \to 0$ as $k \to \infty$. Then $t_k \to -\infty$, since the dominant term $-\frac{a_{11}(1-\phi_k)}{\cos^2 \theta_k} \leq -\frac{a_{11}\delta}{\cos^2 \theta_k} \to -\infty$ while the other two terms grow at most as $O(1/\cos \theta_k)$.

In particular, there exists $K_0 > 0$ such that

$$t_k < -\frac{2a_8}{a_{13}} \quad \text{for all } k \geq K_0 \tag{3.28}$$

where a_{13} is the constant from Lemma 3.15. Since $\hat{B}_{k+1}$ is positive definite, we habe $\text{trace}(\hat{B}_{k+1}) > 0$. Summing (3.27) from $j = K_0$ to k then it holds that

$$\begin{aligned} 0 < \text{trace}(\hat{B}_{k+1}) &\leq \text{trace}(\hat{B}_{K_0}) + a_8(k + 1 - K_0) + \sum_{j=K_0}^{k} t_j \alpha^j \\ &< \text{trace}(\hat{B}_{K_0}) + a_8(k + 1 - K_0) - \frac{2a_8}{a_{13}} \sum_{j=K_0}^{k} \alpha^j \end{aligned} \tag{3.29}$$

On the other hand, by Lemma 3.15, $\prod_{j=1}^{k} \alpha^j \geq a_{13}^k$. Applying the AM–GM inequality to $\alpha^1, \ldots, \alpha^k$

$$\frac{1}{k} \sum_{j=1}^{k} \alpha^j \geq \left(\prod_{j=1}^{k} \alpha^j \right)^{1/k} \geq a_{13}$$

therefore $\sum_{j=1}^{k} \alpha^j \geq k\, a_{13}$, and hence

$$\sum_{j=K_0}^{k} \alpha^j \geq k\, a_{13} - \sum_{j=1}^{K_0 - 1} \alpha^j \tag{3.30}$$

Substituting (3.30) into (3.29) it follows that

$$\begin{aligned} 0 &< \text{trace}(\hat{B}_{K_0}) + a_8(k + 1 - K_0) - \frac{2a_8}{a_{13}} \left(k\, a_{13} - \sum_{j=1}^{K_0 - 1} \alpha^j \right) \\ &= \text{trace}(\hat{B}_{K_0}) + a_8(1 - k - K_0) + \frac{2a_8}{a_{13}} \sum_{j=1}^{K_0 - 1} \alpha^j \end{aligned}$$

The right-hand side tends to $-\infty$ as $k \to \infty$, which is a contradiction. Therefore the assumption $\cos\theta_k \to 0$ is false. It follows that there exist a constant $\delta_0 > 0$ and a subsequence $\{k_j\}$ such that $\cos\theta_{k_j} > \delta_0$ for all j. By Lemma 3.11,

$$f(x^{k_j+1}) - f(x^*) \leq (1 - a_5\delta_0^2)(f(x^{k_j}) - f(x^*))$$

Since $\{f(x^k)\}$ is non-increasing by the Armijo condition and bounded below by $f(x^*)$, the above subsequential contraction implies $f(x^k) \to f(x^*)$, equivalently $x^k \to x^*$. □